BATCH BEFORE/AFTER STUDY

Countermeasure Effectiveness Evaluation Report

18	23.1%	0.702	3	0%
Locations Analyzed	Avg Reduction	Average CMF	Net Prevented	Significant

Report Details

Generated: April 3, 2026

Analysis Radius: 500 ft | Confidence Level: 95%

EPDO Weights: FHWA 2025 (K=883, A=94, B=21, C=11, O=1)

Method: Empirical Bayes (simplified) | Significance: Two-tailed z-test (Poisson)

Executive Summary

Effectiveness Distribution

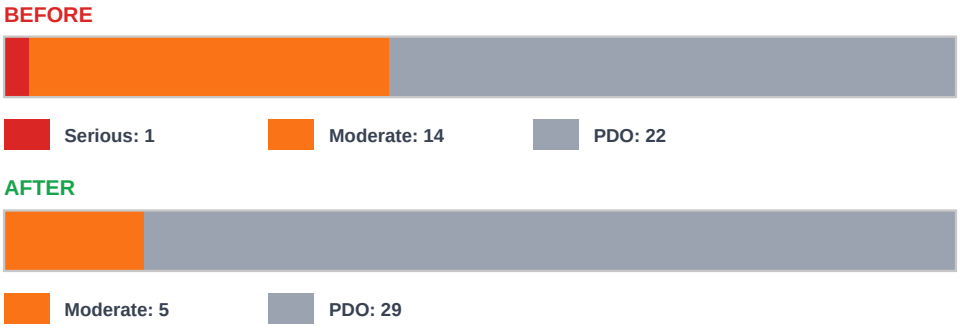
Rating	CMF Range	Count
Highly Effective	CMF < 0.70	6
Effective	CMF 0.70-0.90	1
Marginal	CMF 0.90-1.00	1
Ineffective	CMF 1.00-1.10	5
Negative Impact	CMF > 1.10	1

Statistical Significance

0 of 18 locations (0%) showed statistically significant crash changes at the 95% confidence level.

Net estimated crashes prevented: 3 (sum of before minus after across all locations).

Aggregated Severity: Before vs After



Location Summary Table

Location	Type	Before	After	Change	EPDO B	EPDO A	CMF	Sig.	Rating
Woodman Road Roundabout	Not specified	14	10	**-28.6%**	74	30	0.714	No	**Effective**
University Glendale Henrico	Not specified	1	0	**-100.0%**	1	0	0.000	No	**Highly Effective**
Oak Hill & Doron Roundabout	Not specified	0	2	0.0%	0	22	N/A	No	**N/A**
PH 1 Charles Glen (Charles S	Not specified	0	2	0.0%	0	2	N/A	No	**N/A**
PH 2 Charles Glen (Miami Ave	Not specified	0	0	0.0%	0	0	1.000	No	**Ineffective**
PH 3 Charles Glen (Charles S	Not specified	1	1	0.0%	1	1	0.999	No	**Marginal**
PH 4 Charles Glen (Dartmouth	Not specified	0	1	0.0%	0	1	N/A	No	**N/A**
Colleen & Subrenda Roundabou	Not specified	0	0	0.0%	0	0	1.000	No	**Ineffective**
Turner-Darbytown Modular Rou	Not specified	4	2	**-50.0%**	64	2	0.499	No	**Highly Effective**
Woodman Rd & Magnolia Ridge	Not specified	8	11	**37.5%**	161	31	1.371	No	**Negative Impact**
Francistown at HCMS Permanen	Not specified	4	1	**-75.0%**	64	1	0.250	No	**Highly Effective**
Anoka-Bronwood-Fordson Round	Not specified	0	2	0.0%	0	22	N/A	No	**N/A**
Derbyshire Rd Roundabout at	Not specified	2	1	**-50.0%**	42	21	0.502	No	**Highly Effective**
Twin Hickory Permanent Round	Not specified	2	1	**-50.0%**	2	1	0.495	No	**Highly Effective**
Shady Grove at Old Nuckols R	Not specified	0	0	0.0%	0	0	1.000	No	**Ineffective**
Doran Rd â€“ Four Mile Run P	Not specified	1	0	**-100.0%**	1	0	0.000	No	**Highly Effective**
Neale & Bolling Roundabout	Not specified	0	0	0.0%	0	0	1.000	No	**Ineffective**
Shady Grove at Holman Rd Rou	Not specified	0	0	0.0%	0	0	1.000	No	**Ineffective**

Individual Location Results

1. Woodman Road Roundabout — Not specified

Install Date: 10/19/2021 | Radius: 500 ft | Lat: 37.6715 Lng: -77.4827
Before: 9/18/2017 - 10/18/2021 (4.1 yr) | After: 10/19/2021 - 11/19/2025 (4.1 yr)

Effective

Period	K	A	B	C	O	Unk	Total	EPDO	Rate/Yr
Before	0	0	3	0	11	0	**14**	74	3.4
After	0	0	1	0	9	0	**10**	30	2.4

CMF: 0.714 | CRF: +28.6% | p-value: 0.2840 | Not Significant

2. University Glendale Henrico Roundabout — Not specified

Install Date: 7/19/2022 | Radius: 500 ft | Lat: 37.5920 Lng: -77.5617
Before: 3/18/2019 - 7/18/2022 (3.3 yr) | After: 7/19/2022 - 11/19/2025 (3.3 yr)

Highly Effective

Period	K	A	B	C	O	Unk	Total	EPDO	Rate/Yr
Before	0	0	0	0	1	0	**1**	1	0.3
After	0	0	0	0	0	0	**0**	0	0.0

CMF: 0.000 | CRF: +100.0% | p-value: 0.3171 | Not Significant

3. Oak Hill & Doron Roundabout — Not specified

Install Date: 6/26/2023 | Radius: 500 ft | Lat: 37.5661 Lng: -77.4026
Before: 1/25/2021 - 6/25/2023 (2.4 yr) | After: 6/26/2023 - 11/26/2025 (2.4 yr)

N/A

Period	K	A	B	C	O	Unk	Total	EPDO	Rate/Yr
Before	0	0	0	0	0	0	**0**	0	0.0
After	0	0	1	0	1	0	**2**	22	0.8

CMF: N/A | CRF: N/A | p-value: 1.0000 | Not Significant

Individual Location Results

4. PH 1 Charles Glen (Charles Str & Dartmouth) Rounda

Not specified

Install Date: 7/31/2023 | Radius: 500 ft | Lat: 37.5983 Lng: -77.5229
Before: 3/30/2021 - 7/30/2023 (2.3 yr) | After: 7/31/2023 - 12/1/2025 (2.3 yr)

N/A

Period	K	A	B	C	O	Unk	Total	EPDO	Rate/Yr
Before	0	0	0	0	0	0	**0**	0	0.0
After	0	0	0	0	2	0	**2**	2	0.9

CMF: N/A | CRF: N/A | p-value: 1.0000 | Not Significant

5. PH 2 Charles Glen (Miami Ave & Wakeforest) Roundab

Not specified

Install Date: 8/2/2023 | Radius: 500 ft | Lat: 37.5962 Lng: -77.5227
Before: 5/1/2021 - 8/1/2023 (2.3 yr) | After: 8/2/2023 - 11/2/2025 (2.3 yr)

Ineffective

Period	K	A	B	C	O	Unk	Total	EPDO	Rate/Yr
Before	0	0	0	0	0	0	**0**	0	0.0
After	0	0	0	0	0	0	**0**	0	0.0

CMF: 1.000 | CRF: 0.0% | p-value: 1.0000 | Not Significant

6. PH 3 Charles Glen (Charles Str & Miami) Roundabout

Not specified

Install Date: 8/11/2023 | Radius: 500 ft | Lat: 37.5968 Lng: -77.5240
Before: 5/10/2021 - 8/10/2023 (2.3 yr) | After: 8/11/2023 - 11/11/2025 (2.3 yr)

Marginal

Period	K	A	B	C	O	Unk	Total	EPDO	Rate/Yr
Before	0	0	0	0	1	0	**1**	1	0.4
After	0	0	0	0	1	0	**1**	1	0.4

CMF: 0.999 | CRF: +0.1% | p-value: 0.9990 | Not Significant

Individual Location Results

7. PH 4 Charles Glen (Dartmouth & Wake Forest) Rounda — Not specified

Install Date: 8/25/2023 | Radius: 500 ft | Lat: 37.5974 Lng: -77.5216
Before: 5/24/2021 - 8/24/2023 (2.3 yr) | After: 8/25/2023 - 11/25/2025 (2.3 yr)

N/A

Period	K	A	B	C	O	Unk	Total	EPDO	Rate/Yr
Before	0	0	0	0	0	0	**0**	0	0.0
After	0	0	0	0	1	0	**1**	1	0.4

CMF: N/A | CRF: N/A | p-value: 1.0000 | Not Significant

8. Colleen & Subrenda Roundabout — Not specified

Install Date: 10/5/2023 | Radius: 500 ft | Lat: 37.5211 Lng: -77.3400
Before: 9/4/2021 - 10/4/2023 (2.1 yr) | After: 10/5/2023 - 11/5/2025 (2.1 yr)

Ineffective

Period	K	A	B	C	O	Unk	Total	EPDO	Rate/Yr
Before	0	0	0	0	0	0	**0**	0	0.0
After	0	0	0	0	0	0	**0**	0	0.0

CMF: 1.000 | CRF: 0.0% | p-value: 1.0000 | Not Significant

9. Turner-Darbytown Modular Roundabout — Not specified

Install Date: 11/26/2023 | Radius: 500 ft | Lat: 37.4512 Lng: -77.2994
Before: 11/25/2021 - 11/25/2023 (2.0 yr) | After: 11/26/2023 - 11/26/2025 (2.0 yr)

Highly Effective

Period	K	A	B	C	O	Unk	Total	EPDO	Rate/Yr
Before	0	0	3	0	1	0	**4**	64	2.0
After	0	0	0	0	2	0	**2**	2	1.0

CMF: 0.499 | CRF: +50.1% | p-value: 0.3163 | Not Significant

Individual Location Results

10. Woodman Rd & Magnolia Ridge Roundabout — Not specified

Install Date: 10/31/2023 | Radius: 500 ft | Lat: 37.6741 Lng: -77.4661
Before: 9/30/2021 - 10/30/2023 (2.1 yr) | After: 10/31/2023 - 12/1/2025 (2.1 yr)

Negative Impact

Period	K	A	B	C	O	Unk	Total	EPDO	Rate/Yr
Before	0	1	3	0	4	0	**8**	161	3.8
After	0	0	1	0	10	0	**11**	31	5.3

CMF: 1.371 | CRF: -37.1% | p-value: 0.2930 | Not Significant

11. Francistown at HCMS Permanent Roundabout — Not specified

Install Date: 8/1/2024 | Radius: 500 ft | Lat: 37.6596 Lng: -77.5449
Before: 5/1/2023 - 7/31/2024 (1.3 yr) | After: 8/1/2024 - 11/1/2025 (1.3 yr)

Highly Effective

Period	K	A	B	C	O	Unk	Total	EPDO	Rate/Yr
Before	0	0	3	0	1	0	**4**	64	3.2
After	0	0	0	0	1	0	**1**	1	0.8

CMF: 0.250 | CRF: +75.0% | p-value: 0.1336 | Not Significant

12. Anoka-Bronwood-Fordson Roundabout — Not specified

Install Date: 6/5/2024 | Radius: 500 ft | Lat: 37.6155 Lng: -77.5524
Before: 1/4/2023 - 6/4/2024 (1.4 yr) | After: 6/5/2024 - 11/5/2025 (1.4 yr)

N/A

Period	K	A	B	C	O	Unk	Total	EPDO	Rate/Yr
Before	0	0	0	0	0	0	**0**	0	0.0
After	0	0	1	0	1	0	**2**	22	1.4

CMF: N/A | CRF: N/A | p-value: 1.0000 | Not Significant

Individual Location Results

13. Derbyshire Rd Roundabout at Lakewater Dr — Not specified

Install Date: 9/4/2024 | Radius: 500 ft | Lat: 37.5894 Lng: -77.5925
Before: 7/3/2023 - 9/3/2024 (1.2 yr) | After: 9/4/2024 - 11/4/2025 (1.2 yr)

Highly Effective

Period	K	A	B	C	O	Unk	Total	EPDO	Rate/Yr
Before	0	0	2	0	0	0	**2**	42	1.7
After	0	0	1	0	0	0	**1**	21	0.9

CMF: 0.502 | CRF: +49.8% | p-value: 0.4825 | Not Significant

14. Twin Hickory Permanent Roundabout — Not specified

Install Date: 4/4/2025 | Radius: 500 ft | Lat: 37.6694 Lng: -77.5977
Before: 9/3/2024 - 4/3/2025 (0.6 yr) | After: 4/4/2025 - 11/4/2025 (0.6 yr)

Highly Effective

Period	K	A	B	C	O	Unk	Total	EPDO	Rate/Yr
Before	0	0	0	0	2	0	**2**	2	3.4
After	0	0	0	0	1	0	**1**	1	1.7

CMF: 0.495 | CRF: +50.5% | p-value: 0.4732 | Not Significant

15. Shady Grove at Old Nuckols Rd Roundabout — Not specified

Install Date: 4/4/2025 | Radius: 500 ft | Lat: 37.6844 Lng: -77.5982
Before: 9/3/2024 - 4/3/2025 (0.6 yr) | After: 4/4/2025 - 11/4/2025 (0.6 yr)

Ineffective

Period	K	A	B	C	O	Unk	Total	EPDO	Rate/Yr
Before	0	0	0	0	0	0	**0**	0	0.0
After	0	0	0	0	0	0	**0**	0	0.0

CMF: 1.000 | CRF: 0.0% | p-value: 1.0000 | Not Significant

Individual Location Results

16. Doran Rd " Four Mile Run PKWY " Danrett Ln Roundab

Not specified

Install Date: 6/24/2025 | Radius: 500 ft | Lat: 37.4523 Lng: -77.3330
Before: 2/23/2025 - 6/23/2025 (0.3 yr) | After: 6/24/2025 - 10/24/2025 (0.3 yr)

Highly Effective

Period	K	A	B	C	O	Unk	Total	EPDO	Rate/Yr
Before	0	0	0	0	1	0	**1**	1	3.0
After	0	0	0	0	0	0	**0**	0	0.0

CMF: 0.000 | CRF: +100.0% | p-value: 0.3132 | Not Significant

17. Neale & Bolling Roundabout

Not specified

Install Date: 12/19/2025 | Radius: 500 ft | Lat: 37.5784 Lng: -77.3883
Before: 9/3/2025 - 12/18/2025 (0.3 yr) | After: 12/19/2025 - 4/3/2026 (0.3 yr)

Ineffective

Period	K	A	B	C	O	Unk	Total	EPDO	Rate/Yr
Before	0	0	0	0	0	0	**0**	0	0.0
After	0	0	0	0	0	0	**0**	0	0.0

CMF: 1.000 | CRF: 0.0% | p-value: 1.0000 | Not Significant

18. Shady Grove at Holman Rd Roundabout

Not specified

Install Date: 1/16/2026 | Radius: 500 ft | Lat: 37.6944 Lng: -77.5928
Before: 10/29/2025 - 1/15/2026 (0.2 yr) | After: 1/16/2026 - 4/3/2026 (0.2 yr)

Ineffective

Period	K	A	B	C	O	Unk	Total	EPDO	Rate/Yr
Before	0	0	0	0	0	0	**0**	0	0.0
After	0	0	0	0	0	0	**0**	0	0.0

CMF: 1.000 | CRF: 0.0% | p-value: 1.0000 | Not Significant

Methodology Notes

Analysis Method

Empirical Bayes (simplified) with Poisson variance approximation. The expected after-period crash count is estimated by adjusting the before-period count for the ratio of study period lengths.

Statistical Significance

Two-tailed z-test based on Poisson distribution. Confidence level: 95%.
A location is flagged as significant when p-value < 0.05.

Crash Modification Factor (CMF)

CMF = Observed After-Period Crashes / Expected After-Period Crashes.
Values less than 1.0 indicate crash reduction. Values greater than 1.0 indicate crash increase.

Crash Reduction Factor (CRF)

CRF = (1 - CMF) x 100. Positive values = crash reduction percentage.

EPDO (Equivalent Property Damage Only)

Weights: FHWA 2025
K = 883, A = 94, B = 21, C = 11, O = 1
Source: FHWA-SA-25-021 (Oct 2025) crash costs in 2024 dollars

Effectiveness Ratings

Highly Effective:	CMF < 0.70 (greater than 30% crash reduction)
Effective:	CMF 0.70 - 0.90 (10-30% reduction)
Marginal:	CMF 0.90 - 1.00 (0-10% reduction)
Ineffective:	CMF 1.00 - 1.10 (0-10% increase)
Negative Impact:	CMF > 1.10 (greater than 10% increase)

Limitations

This analysis uses a simplified EB method that adjusts for period length but does not incorporate Safety Performance Functions (SPFs) or reference group data. For HSIP-grade documentation, use the full single-location Before/After Study tab with complete EB methodology.